



Antibody dependent enhancement: Unavoidable problems in vaccine development

[Lele Xu](#), [Zhiqian Ma](#), [Yang Li](#), [Zhaoxia Pang](#), and [Shuqi Xiao](#)*

► [Author information](#) ► [Copyright and License
information](#) [PMC Disclaimer](#)

Abstract

[Go to:](#) ►

In some cases, antibodies can enhance virus entry and replication in cells. This phenomenon is called antibody-dependent infection enhancement (ADE). ADE not only promotes the virus to be recognized by the target cell and enters the target cell, but also affects the signal transmission in the target cell. Early formalin-inactivated virus vaccines such as aluminum adjuvants (RSV and measles) have been shown to induce ADE. Although there is no direct evidence that there is ADE in COVID-19, this potential risk is a huge challenge for prevention and vaccine development. This article focuses on the virus-induced ADE phenomenon and its molecular mechanism. It also summarizes various attempts in vaccine research and development to eliminate the ADE phenomenon, and proposes to avoid A

Feedback



1. Introduction

[Go to: ►](#)

In 1967 Hawkes first confirmed that IgG in serum can induce ADE ([Lafferty, 1967](#)). Many viruses have been proven to have ADE effects, such as *Arthropod borne viruses* (ABV), *Dengue virus* (DENV), *Respiratory syncytial virus* (RSV), *Human Immunodeficiency virus* (HIV), *Feline infectious peritonitis virus* (FIPV), *Coronaviruses* (CoV). For many viruses (including DENV, HIV) that pose a major threat to human health, the presence of ADE is considered to be a major obstacle to vaccine development.

Fc receptor (FcR)-mediated ADE is the most common form of ADE, which was first discovered by [Halstead \(1977\)](#). At first, it was thought that the antigen antibody immune complex formed by virus and specific antibody combined with the host cell with the help of FcR on the cell surface, which was more conducive to the entry of virus, and increased the infection amount and infection rate of virus, finally led to the increase of infection and replication of virus. By phagocytosing immune complexes, the cells expressing FcR on the surface, such as monocytes, macrophages, dendritic cells, and certain granulocytes, can produce ADE. This kind of ADE is main

by IgG antibody, but IgM, IgE and IgA

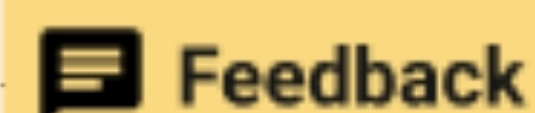
Feedback



([Walls et al., 2019](#)). Therefore, SARS-CoV is likely to have ADE mechanism similar to MERS-CoV.

In the acidic pH environment of the endosome leads to the aggravation of antibody mediated infection, which is the opposite of SARS-CoV infection mediated by ACE2 receptor. Virus particles that infect cells via ADE pathway may still be trapped in acid compartment, and inhibition of internal acidification will prevent its degradation. It was also found that the sera induced by ADE did not contain anti-Spike IgG2a antibody, while the two neutralized/non-enhanced sera did ([Jaume et al., 2011](#)). IgG2a is a marker of Th1 type response, and Th2 type is more prone to induce ADE. Whether this phenomenon is related to the preference for Th1 type response and whether the deficiency of IgG2a has a causal relationship with the occurrence of ADE infection needs further research.

Daniel found two VHH (variable domain of the heavy-chain of heavy chain antibody) targeting SARS-CoV and MERS-CoV RBD, which neutralizes SARS-CoV and MERS-CoV and interferes with receptor binding ([Wrapp et al., 2020](#)). The possible mechanisms of neutralization are the blocking of the receptor binding inter



I soon stumbled upon another immunological conundrum in dengue that goes deeper than the skin. I discovered that the child's risk of severe dengue would be higher if they were reinfected in the future.

This risk would be further heightened if the reinfection were caused by a different version of the same virus, of which four distinct types exist.

Once again, I was faced with a contradiction. This increased risk of severe dengue during a secondary infection is counterintuitive to a central tenet of our immune system: memory. In theory, previous infections should train our immune system to respond better and faster and to have a stronger response to a familiar virus, but that was not always the case for dengue.

Additionally, some infants, whose mothers had contracted dengue fever during pregnancy, also had a greater risk of developing severe dengue. These observations implicated antibodies as a possible cause. Virus-specific antibodies obtained either from a previous infection or from the mother through placental transfer could worsen dengue through a phenomenon known as antibody-dependent enhancement (2).

At present, not much is known about why antibody-dependent enhancement only occurs in some patients and under specific circumstances depending on the type, concentration, and quality of circulating antibodies. In fact, the world's first dengue vaccine acted as a jarring reminder that vaccine-induced antibodies are also capable of disease enhancement.

During its mass rollout in the Philippines in 2016, the



During its mass rollout in the Philippines in 2016, the risk of hospitalization and death from a subsequent infection after vaccination was higher in children under nine years of age (3), who likely have never had a natural infection. There has been an understanding regarding antibody-dependent enhancement in dengue since 1970 (4); given this, how was this adverse outcome not better predicted? As a result, not only were some children deprived of a vaccine that could prevent dengue-related deaths in the region, but general efforts to improve vaccine acceptance and eradicate vaccine-preventable diseases were also compromised. Sales and distribution of the dengue vaccine were halted, but the damage was already done. Vaccine confidence in Southeast Asia plummeted (5).

The challenges of antibody-dependent enhancement resurfaced during the COVID-19 pandemic, as predicting this phenomenon in new diseases can be difficult with our incomplete understanding. Amid debate about its occurrence in COVID-19, scientists at the Jenner Institute ran additional safety assessments while developing the Oxford-AstraZeneca COVID-19 vaccine, including preclinical testing in primates to ensure that disease enhancement did not occur with COVID-19 vaccine-induced antibodies (6). Besides these costly and time-consuming vaccine safety measures, uncertainties about antibody-dependent enhancement in COVID-19 also restricted the clinical



Table of Contents

Potential Risks of Antibodies

Antibodies provide many benefits. They eliminate infections and provide protective immunity against future infections.

However, in rare circumstances, antibodies might actually worsen an infection. For example, antibodies might bind to a virus in such a way that makes it easier to enter cells.

This might mean that if a person is re-infected after an initial mild infection, they could then have a more severe case the second time. Or, it *theoretically* might mean that a person could have a worse response to potential infection if they have previously been vaccinated for the disease.

This scenario has been called “antibody-dependent enhancement.”^[2] It has been found in viruses such as [dengue](#). In that virus, it complicated the creation of successful vaccines.

Because researchers were aware of this theoretical possibility, they have been looking very carefully to see whether this might be possible in COVID-19.

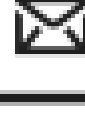
Ad

Kids aren't thinking about HPV cancers, that's why you need to. [Learn more](#)



Review

Post-vaccination encephalomyelitis: Literature review and illustrative case

William Huynh^a, Dennis J. Cordato^b  , Elias Kehdi^a, Lynette T. Masters^c,
Chris Dedousis^d

[Show more](#) [+](#) Add to Mendeley  Share  Cite<https://doi.org/10.1016/j.jocn.2008.05.002> [Get rights and content](#) 

Abstract

Acute disseminated encephalomyelitis (ADEM) is an inflammatory demyelinating disease of the central nervous system that is usually considered a monophasic disease. ADEM forms one of several categories of primary inflammatory demyelinating disorders of the central nervous system including multiple sclerosis, optic neuropathy, acute transverse myelitis, and neuromyelitis optica (Devic's disease). Post-infectious and post-immunisation encephalomyelitis make up about three-quarters of cases, where the timing of a febrile event is associated with the onset of neurological disease. Post-vaccination ADEM has been associated with several vaccines such as rabies, diphtheria–tetanus–polio, smallpox, measles, mumps, rubella, Japanese B encephalitis, pertussis, influenza, hepatitis B, and the Hog vaccine. We review ADEM with particular emphasis on vaccination as the precipitating factor. We performed a literature search using Medline (1976–2007) with search terms including “ADEM”, “acute disseminated encephalomyelitis”, “encephalomyelitis”, “vaccination”, and “immunisation”. A patient presenting with bilateral optic neuropathies within 3 weeks of “inactivated” influenza vaccination followed by delayed onset of ADEM 3 months post-vaccination is described.

ADVERSE EVENTS





Case Series Analysis of Adverse Reactions to Vaccines: A Comparative Evaluation

C. P. Farrington,¹ J. Nash,² and E. Miller³

A modified cohort method has been proposed for estimating the relative incidence of rare adverse reactions after vaccination. The method requires only a sample of the cases, thus avoiding the need for following large population cohorts or selecting controls. This case series method has statistical power equivalent to that of the full cohort method when the risk periods after vaccination are short and vaccine coverage is high. The method also eliminates confounding by variables associated with both the outcome and avoidance of vaccination. In this paper, the cohort, case-control, and case series methods are reviewed, and their underlying assumptions and performances are compared. Theoretical results are illustrated using data on febrile convulsions after measles-mumps-rubella vaccination in the United Kingdom. *Am J Epidemiol* 1996;143:1165–73.

case-control studies; cohort studies; epidemiologic methods; statistics; vaccination

Vaccine safety with respect to rare adverse events is an issue of substantial public and epidemiologic interest. However, while safety with respect to common reactions may be established in clinical trials, these studies do not usually possess sufficient power for testing hypotheses of association with rare adverse events. Epidemiologic studies are therefore required to investigate suspected or hypothesized associations once the vaccine is in widespread use.

Such studies present several methodological difficulties. Many potential sources of bias have been identified, such as differential ascertainment of cases in recently vaccinated and unvaccinated individuals and differential vaccination rates in individuals at higher or lower risk (1). Cohort studies must be very large to achieve sufficient power, and may therefore be impractical, though it is sometimes possible to undertake retrospective cohort studies in large data sets assembled for other purposes (2). Case-control studies require much smaller overall sample sizes, but are open to bias in the selection of

controls. Both cohort and case-control studies may suffer from confounding by variables related both to avoidance of vaccination and to the outcome of interest. Thus, for instance, Fine and Chen (1) list parental education, ethnic group, age of the mother, maternal smoking, birth weight, evolving neurologic disorders, and conditions predisposing to seizures as factors related both to vaccination and to sudden infant death syndrome or encephalopathy. Few studies could realistically aspire to control for all of these factors.

An alternative study design using only data on cases has been proposed for investigating acute adverse events (3). This design combines the power and simplicity of the cohort method and the economy of the case-control method, while reducing confounding. The method is similar in spirit to the case-crossover design proposed by Maclure (4) for studying the effect of transient exposures on myocardial infarction and to that described by Feldman (5) for assessing drug reactions. It differs from these methods in that it is derived from the same statistical model as the cohort study design, from which it inherits many properties, including the facility for handling several dependent exposures and controlling for age effects, both of which are essential in the present context. It also avoids the need to specify the prior probabilities of exposure (6). The case series method has been used to estimate the relative incidence of febrile convulsions after diphtheria-tetanus-pertussis (DTP) and measles-mumps-rubella (MMR) vaccination in the United Kingdom (7).

Received for publication March 30, 1995, and in final form February 26, 1996.

Abbreviations: DTP, diphtheria-tetanus-pertussis; MMR, measles-mumps-rubella.

¹ PHLS Statistics Unit, Public Health Laboratory Service, London NW9 5EQ, England.

² Public Health Laboratory, William Harvey Hospital, Ashford, Kent TN24 0LZ, England.

³ Immunization Division, PHLS Communicable Disease Surveillance Centre, Public Health Laboratory Service, London NW9 5EQ, England.

Reprint requests to Dr. C. P. Farrington, PHLS Statistics Unit, Public Health Laboratory Service, 61 Colindale Avenue, London NW9 5EQ, England.

For example, FDA believes that its system for gathering information about ADEs, the Adverse Event Reporting System (AERS), receives reports for only about **1 to 10 percent** of all ADEs.



Government Accountability Office (GAO) (.gov)

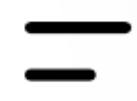
<https://www.gao.gov> › [assets](#) › [t-hehs-00-61](#)



Surveillance Systems for Adverse Events and Medical Errors

About featured snippets • Feedback

9:47



ChatGPT 4o >



Calculation:

$$\text{Estimated True Total} = \text{Reported Events} \times 100 =$$

So, if only 1% of adverse events have been reported, the estimated true total would be **185,560,100 adverse events** since 2020.

What percent of the American population would that be?



To calculate what percent of the American population 185,560,100 adverse events would represent, we need to know the current population of the United States. As of 2024, the U.S. population is estimated to be around 335 million.

Calculation:

$$\text{Percentage of Population} = \left(\frac{\text{Estimated True Total}}{\text{Total U.S. Population}} \right) \times 100$$

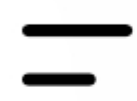
$$\text{Percentage of Population} = \left(\frac{185,560,100}{335,000,000} \right) \times 100$$

So, **185,560,100 adverse events** would represent approximately **55.38%** of the American population.



Message





So, **185,560,100 adverse events** would represent approximately **80.68%** of the vaccinated population in the United States.

If 10% were reported instead of 1%, how would that number change?



If 10% of adverse events were reported instead of 1%, then the reported number would represent a larger portion of the total, so the estimated true total would be lower.

Given:

- **Reported adverse events:** 1,855,601
- **Reported percentage:** 10%

Calculation:

To find the true total if 10% were reported, you would multiply the reported number by 10 instead of 100.

$$\text{Estimated True Total} = \text{Reported Events} \times \frac{100}{10} =$$

So, if 10% of adverse events were reported, the estimated true total would be **18,556,010 adverse events**.



Message



From the 7/26/2024 release of VAERS data:
Using all 2,618,691 Cases in VAERS
[Government Disclaimer on use of this data](#)

Graph

Count by Year of Appearance in VAERS

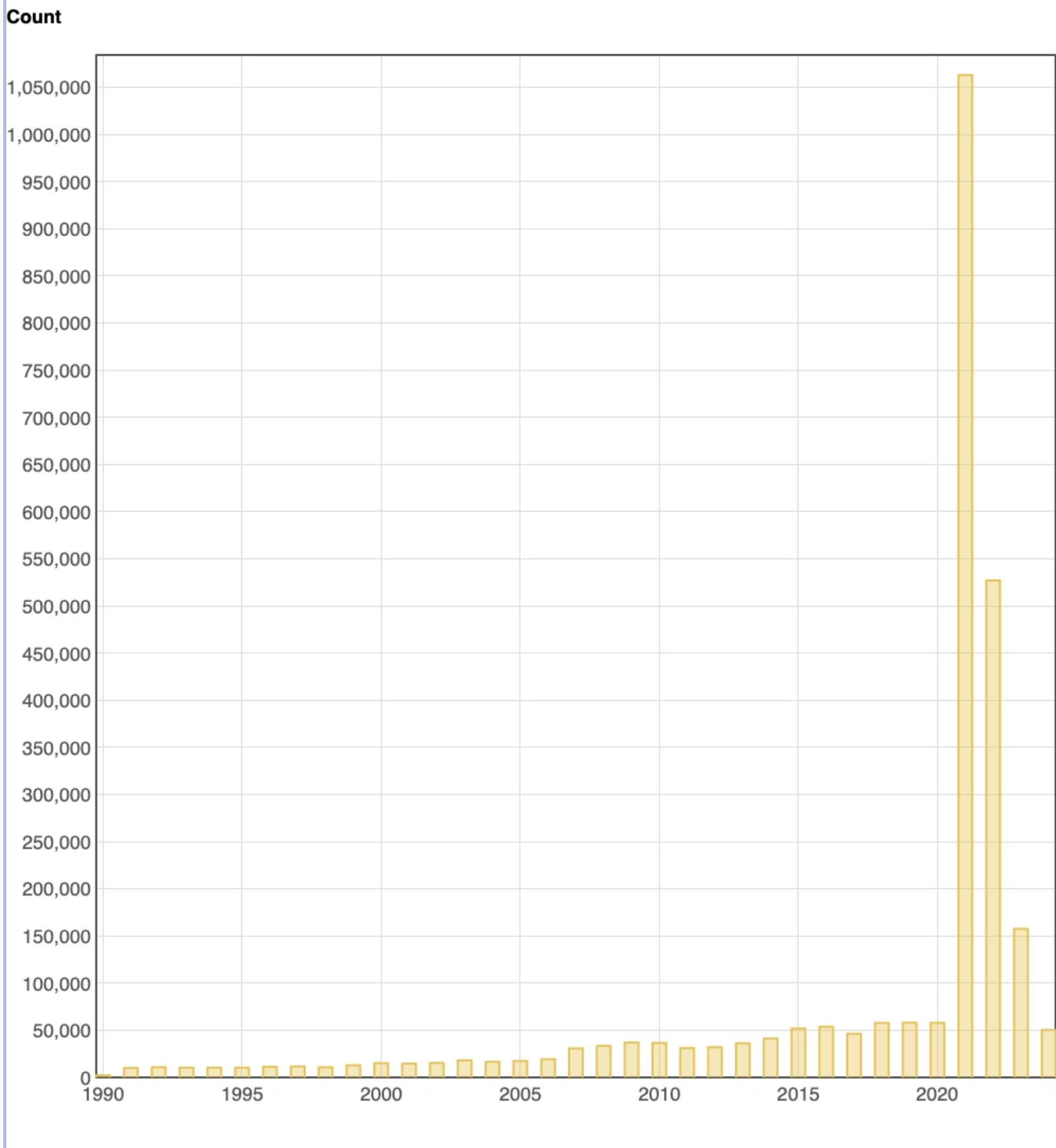


Chart type: Bar ☒ Swap Axes ☐ Show Percent ☐ Stack Bars

Width (pixels): Height (pixels): Resize

Zoom:

From the 7/26/2024 release of VAERS data:

Using all 2,618,691 Cases in VAERS

[Government Disclaimer on use of this data](#)

Table		
↓ Year of Appearance in VAERS	↑ ↓ Count	Percent
1990	2,151	0.08%
1991	9,992	0.38%
1992	10,817	0.41%
1993	10,308	0.39%
1994	10,354	0.4%
1995	10,273	0.39%
1996	11,189	0.43%
1997	11,608	0.44%
1998	10,782	0.41%
1999	12,878	0.49%
2000	15,112	0.58%
2001	14,633	0.56%
2002	15,331	0.59%
2003	18,077	0.69%
2004	16,506	0.63%
2005	17,447	0.67%
2006	19,330	0.74%
2007	30,813	1.18%
2008	33,443	1.28%
2009	37,042	1.41%
2010	36,576	1.4%
2011	31,126	1.19%
2012	32,107	1.23%
2013	36,245	1.38%
2014	41,207	1.57%
2015	51,933	1.98%
2016	53,695	2.05%
2017	46,324	1.77%
2018	57,789	2.21%
2019	58,002	2.21%
2020	57,779	2.21%
2021	1,062,925	40.59%
2022	527,053	20.13%
2023	157,602	6.02%
2024	50,242	1.92%
TOTAL	2,618,691	100%



< Previous Article Next Article >

Understanding the Role of Human Variation in Vaccine Adverse Events: The Clinical Immunization Safety Assessment Network

Philip S. LaRussa, MD ; Kathryn M. Edwards, MD; Cornelia L. Dekker, MD; Nicola P. Klein, MD, PhD; Neal A. Halsey, MD; Colin Marchant, MD; Roger Baxter, MD; Renata J. M. Engler, MD; Jennifer Kissner, PhD; Barbara A. Slade, MD, MS

Address correspondence to Philip LaRussa, MD, Professor of Clinical Pediatrics, Columbia. E-mail: plarussa@columbia.edu

FINANCIAL DISCLOSURE: Dr LaRussa receives contract funding from the National Institutes of Health (NIH) and the CDC and currently serves on a Novartis data safety monitoring board; Dr Edwards receives contract funding from the NIH and the CDC, has received contract support within the past 3 years from Sanofi Pasteur, Novartis, Wyeth, and CSL for the conduct of clinical vaccine studies, and has served as a consultant to Nexbio and PATH; Dr Dekker receives contract funding from the NIH and the CDC; Dr Klein receives contract funding from the CDC and research support from GlaxoSmithKline, Merck & Co, Sanofi Pasteur, Wyeth (Pfizer), Novartis and MedImmune; Dr Halsey receives contract funding from the CDC and is compensated for serving on safety monitoring committees for studies of vaccines for Merck and Novartis, has received an honorarium for training and support for travel from Sanofi-MSD, a company in France, has research grants through Johns Hopkins University for studies of unrelated vaccines in Guatemala from Berna and Intercel, and has received support for a study of a Merck HPV vaccine in Peru but receives no support for the effort in Peru; Dr Marchant receives contract funding from the CDC and has performed clinical research for GlaxoSmithKline, Sanofi Pasteur, Merck, Wyeth (Pfizer Inc), MedImmune, and Novartis, has served as a consultant to GlaxoSmithKline, Sanofi Pasteur, MedImmune, and Novartis, and has given lectures sponsored by GlaxoSmithKline and Sanofi Pasteur; Dr Baxter receives contract funding from the CDC and research grants from Sanofi Pasteur, MedImmune, Novartis, GlaxoSmithKline, Pfizer, Protein Sciences, and Merck. Drs Engler, Kissner, and Slade have indicated they have no financial relationships relevant to this article to disclose.

“IT’S NOT THE VACCINES FAULT,
IT’S JUST BAD HUMAN DESIGN WITH SOME PEOPLE”

Take Note - These Are Adverse Events Possible After Any And Every Immunization.

2. Detecting and reporting AEFIs

An adverse event following immunization is a medical incident that takes place after an immunization and is believed to be caused by the immunization. Although people often think that a medical incident after an immunization must be caused by the immunization, many such incidents are coincidental. Another belief - that vaccine is the most common cause of AEFIs - is also mistaken. Programme error, which can be prevented, is more often the cause.

To find the cause of an AEFI, the events must first be detected and reported. Countries may not have the resources to include all AEFIs in their surveillance systems, and it is not necessary to do so. Managers have choices in terms of which ones to report, how they are reported, what is reported, and when. In this chapter, these choices are discussed and recommended practices examined.

2.1 Which AEFIs should be included in AEFI surveillance?

All immunization programmes should monitor at least the following AEFIs (see Appendix A for case definitions):

- (1) All injection site abscesses.
- (2) All cases of BCG lymphadenitis
- (3) All deaths that are thought by health workers, or the public, to be related to immunization.
- (4) All cases requiring hospitalization that are thought by health workers, or the public, to be related to immunization.
- (5) Other severe or unusual medical incidents that are thought by health workers, or the public, to be related to immunization.

With respect to the third, fourth, and fifth events, health workers may relate the event to immunization because it occurred within a month of an immunization, as its case definition indicates. However, some medical incidents can be related to immunization even if they have a delayed onset. The case definitions in Appendix A describe the most common of these incidents.

The above five categories of AEFIs are sometimes called "trigger" events because their presence stimulates or triggers a response.

2. Central nervous system adverse events

Acute Paralysis

Vaccine-Associated Paralytic Poliomyelitis:

- Acute onset of flaccid paralysis within 4 - 30 days of receipt of oral polio-virus vaccine (OPV), or within 4 -75 days after contact with a vaccine recipient, with neurological deficits remaining 60 days after onset, or death.
- Guillain-Barré Syndrome (GBS): Acute onset of rapidly progressive, ascending, symmetrical flaccid paralysis, without fever at onset of paralysis and with sensory loss. Cases are diagnosed by cerebrospinal fluid (CSF) investigation showing dissociation between cellular count and protein content.

GBS occurring within 30 days after immunization should be reported.

Encephalopathy

Encephalopathy is an acute onset of major illness temporally linked with immunization and characterized by any two of the following three conditions:

- Seizures;
- Severe alteration in level of consciousness lasting for one day or more; and
- Distinct change in behaviour lasting one day or more.

Cases occurring within 72 hours after vaccination should be reported.

Encephalitis

Encephalitis is characterized by the above mentioned symptoms and signs of cerebral inflammation and, in many cases, CSF pleocytosis and/or virus isolation.

Any encephalitis occurring within 1 - 4 weeks following immunization should be reported.

Meningitis

Acute onset of major illness with fever, neck stiffness/positive meningeal signs (Kernig, Brudzinski). Symptoms may be subtle to similar to those of encephalitis. CSF examination is the most important diagnostic measure: CSF pleocytosis and/or detection of microorganism (Gram stain or isolation).

Seizures

Seizures lasting from several minutes to more than 15 minutes and not accompanied by focal neurological signs or symptoms.

- Febrile Seizures; or
- Afebrile Seizures.

3. Other adverse events

Allergic reaction

Characterized by one or more of the following: (1) skin manifestations (e.g. hives, eczema); (2) wheezing; (3) facial or generalized oedema.

Anaphylactoid Reaction (acute hypersensitivity reaction)

Exaggerated acute reaction, occurring within 2 hours after immunization, characterized by one or more of the following: (1) wheezing and shortness of breath due to bronchospasm; (2) laryngospasm/laryngeal oedema; (3) one or more skin manifestations, e.g. hives, facial oedema, or generalized oedema.

Anaphylactic Shock

Circulatory failure (e.g. alteration of the level of consciousness, low arterial blood pressure, weakness or absence of peripheral pulses, cold extremities secondary to reduced peripheral circulation, flushed face and increased perspiration) with or without bronchospasm and/or laryngospasm/laryngeal oedema leading to respiratory distress occurring immediately after immunization.

Arthralgia

Joint pain usually including the small peripheral joints.

- Persistent: Joint pain lasting longer than 10 days.
- Transient: Joint pain lasting up to approximately 10 days.

Disseminated BCG-itis

Disseminated infection occurring within 1 to 12 months after BCG vaccination and confirmed by isolation of *Mycobacterium bovis* BCG strain.

Fever

- Fever, mild: Temperature 38°C to 38.9°C (rectal)
- Fever, high: Temperature 39°C to 40.4°C (rectal)
- Fever, extreme (hyperpyrexia): Temperature higher than or equal to 40.5°C (rectal)
- Fever, unspecified: Temperature presumed to be high but not measured.

Only high and extreme fever should be reported.

Hypotensive-Hyporesponsive Episode (shock collapse)

Sudden onset of paleness, decreased level or loss of responsiveness, decreased level or loss of muscle tone (occurring within 24 hours of vaccination). The episode is transient and self-limiting.

Osteitis/Osteomyelitis

Inflammation of the bone either due to BCG immunization (occurring within 8 to 16 months after immunization) or caused by other bacterial infection.

Persistent Screaming

Inconsolable continuous crying lasting at least 3 hours accompanied by high-pitched screaming.

Sepsis

Acute onset of severe generalized illness due to bacterial infection and confirmed by positive blood culture.

Toxic-Shock Syndrome

Abrupt onset of fever, vomiting and watery diarrhea within a few hours of immunization, often leading to death within 24-48 hours.

4. Other severe and unusual events occurring within 4 weeks after immunization and not covered under no. 1, 2, or 3.

Any death of a vaccine recipient temporally linked (within 4 weeks) to immunization, where no other clear cause of death can be established, should be reported.

In addition, any unusual events should be reported.

I follow subreddits like newparents, afterthebump, etc. and cannot explain to you how many moms are going through hell because their babies will not stop screaming crying, all describe it as a painful, uncomfortable cry. Many describe their babies arching their back when they are laid down and sudden GERD and diarrhea issues. For some reason, many of these posts are from moms of 4 month or 6 month olds.

TABLE 5. Vaccine Adverse Event Reporting System (VAERS) reports of frequently reported vaccines or vaccine combinations* — United States, 1991–1995

Vaccines or vaccine combinations	No.	(%†)	Total %
HEP	12,519	(24.2)	24.2
DTP HibV OPV	5,344	(10.3)	34.5
FLU	4,696	(9.1)	43.5
MMR	3,386	(6.5)	50.1
TD	2,510	(4.8)	54.9
DTP OPV	2,300	(4.4)	59.4
DTP	1,595	(3.1)	62.4
DTP HibV	1,564	(3.0)	65.5
DTP HEP HibV OPV	1,543	(3.0)	68.4
DTP MMR OPV	1,523	(2.9)	71.4
DTP HibV MMR OPV	1,429	(2.8)	74.1
PPV	974	(1.9)	76.0
HibV MMR	880	(1.7)	77.7
DTPH OPV	877	(1.7)	79.4
DTPH HEP OPV	784	(1.5)	80.9
HibV	722	(1.4)	82.3
RAB	646	(1.2)	83.6
VARCEL	574	(1.1)	84.7
FLU PPV	455	(0.9)	85.5
TYP	408	(0.8)	86.3
DT	397	(0.8)	87.1
MMR TD	395	(0.8)	87.9
TTOX	384	(0.7)	88.6
DTPH	375	(0.7)	89.3
R	306	(0.6)	89.9
DTAP MMR OPV	265	(0.5)	90.4
DTAP OPV	227	(0.4)	90.9
DTP HibV MMR	218	(0.4)	91.3
DTP HEP HibV	215	(0.4)	91.7
M	197	(0.4)	92.1
DTP MMR	178	(0.3)	92.4
DTPH MMR OPV	156	(0.3)	92.7
DTAP HibV MMR OPV	152	(0.3)	93.0
DTPH MMR	141	(0.3)	93.3
HEP MMR	128	(0.2)	93.5
HibV MMR OPV	120	(0.2)	93.8
DTAP	117	(0.2)	94.0
MMR OPV TD	101	(0.2)	94.2
DTP HEP OPV	100	(0.2)	94.4
DTPH HEP	92	(0.2)	94.6
DTP HEP HibV MMR OPV	89	(0.2)	94.7
HibV OPV	89	(0.2)	94.9
MMR OPV	80	(0.2)	95.1
HEP TD	72	(0.1)	95.2
DT OPV	68	(0.1)	95.3
DT MMR OPV	67	(0.1)	95.5
DT MMR	66	(0.1)	95.6
FLU TD	63	(0.1)	95.7
MR	63	(0.1)	95.8
OPV	63	(0.1)	96.0
HEPA	61	(0.1)	96.1
YF	58	(0.1)	96.2
HEP HibV OPV	57	(0.1)	96.3
DTP HibV IPV	52	(0.1)	96.4
OPV TD	52	(0.1)	96.5
TD YF	50	(<0.1)	96.6
Other‡	1,765	(3.4)	100.0

* The frequently reported vaccines or vaccine combinations were defined as the vaccines or vaccine combinations for which ≥50 reports were received during 1991–1995.
† Percentage represents the proportion of reports that include the vaccine.

Original Article

Analysis of health outcomes in vaccinated and unvaccinated children: Developmental delays, asthma, ear infections and gastrointestinal disorders

Brian S Hooker ¹ and Neil Z Miller²

Objective: The aim of this study was to compare the health of vaccinated versus unvaccinated pediatric populations.

Methods: Using data from three medical practices in the United States with children born between November 2005 and June 2015, vaccinated children were compared to unvaccinated children during the first year of life for later incidence of developmental delays, asthma, ear infections and gastrointestinal disorders. All diagnoses utilized International Classification of Diseases–9 and International Classification of Diseases–10 codes through medical chart review. Subjects were a minimum of 3 years of age, stratified based on medical practice, year of birth and gender and compared using a logistic regression model.

Results: Vaccination before 1 year of age was associated with increased odds of developmental delays (OR = 2.18, 95% CI 1.47–3.24), asthma (OR = 4.49, 95% CI 2.04–9.88) and ear infections (OR = 2.13, 95% CI 1.63–2.78). In a quartile analysis, subjects were grouped by number of vaccine doses received in the first year of life. Higher odds ratios were observed in Quartiles 3 and 4 (where more vaccine doses were received) for all four health conditions considered, as compared to Quartile 1. In a temporal analysis, developmental delays showed a linear increase as the age cut-offs increased from 6 to 12 to 18 to 24 months of age (ORs = 1.95, 2.18, 2.92 and 3.51, respectively). Slightly higher ORs were also observed for all four health conditions when time permitted for a diagnosis was extended from ≥ 3 years of age to ≥ 5 years of age.

Conclusion: In this study, which only allowed for the calculation of unadjusted observational associations, higher ORs were observed within the vaccinated versus unvaccinated group for developmental delays, asthma and ear infections. Further study is necessary to understand the full spectrum of health effects associated with childhood vaccination.

Institutional Review Board Guidebook

* CHAPTER V *

BIOMEDICAL AND BEHAVIORAL RESEARCH: AN OVERVIEW

A. Introduction	D. Medical Devices
Biomedical Research	E. Use of Radioactive Materials and X-Rays
Behavioral Research	F. AIDS/HIV-Related Research
Fieldwork	G. Transplants
Social Policy Experimentation	H. Human Genetic Research
B. Drug Trials	I. Alcohol and Drug Research
C. Vaccine Trials	Suggestions for Further Reading

A. INTRODUCTION

Most of the research reviewed by IRBs falls within the broad categories of biomedical or behavioral research. IRBs should be sensitive to specific aspects of biomedical and behavioral research in their review of protocols.

C. VACCINE TRIALS

INTRODUCTION

Vaccines are used to prevent infectious diseases. Successful vaccine trials have resulted in the development of safe and effective vaccines for polio, measles, rubella, hepatitis B, pneumococcal pneumonia, and other serious diseases. Currently, vaccines are being evaluated to prevent infectious diseases such as AIDS (or transmission of HIV), malaria, tuberculosis, trachoma, cytomegalovirus, herpes simplex, and influenza. Vaccines must undergo clinical testing prior to approval and licensure by the **FDA**. The regulations governing the conduct of clinical trials on investigational vaccines are the same as those governing the conduct of **investigational new drug** research [*see* Guidebook Chapter 5, Section B, "Drug Trials"]; however, the **risks** and **benefits** associated with vaccine trials may differ from those of drug trials.

A vaccine is a biologic; its use in trials involving human subjects is similar to the use of any drug. Vaccines do, however, differ from therapeutic drugs in two important ways. As used here, they are not designed to diagnose or cure disease in afflicted individuals; their purpose is to prevent a particular disease in healthy human beings. Vaccines are also used to protect people with a high statistical risk for contracting a particular disease or for suffering especially serious consequences from a disease. Vaccines trigger the body's normal immune response, producing antibodies that protect against future infection. Some vaccines (*e.g.*, those containing active microorganisms or live-attenuated vaccines) have a small but real disease-producing capacity. Thus, one rare risk of a new vaccine is the possibility of infecting a healthy subject with the very disease researchers are seeking to prevent. More often, however, subjects involved in vaccine trials temporarily suffer from some of the symptoms and effects of the disease (*e.g.*, polio, German measles) as they acquire immunity.

DEFINITIONS

- **Biologic:** Any therapeutic serum, toxin, anti-toxin, or analogous microbial product applicable to the prevention, treatment, or cure of diseases or injuries.
- **Purity:** The relative absence of extraneous matter in a vaccine that may or may not be harmful to the recipient or deleterious to the product.
- **Sterility:** The absence of viable contaminating microorganisms; aseptic state.
- **Vaccine:** A biologic product generally made from an infectious agent or its components — a virus, bacterium or other microorganism — that is killed (inactive) or live-attenuated (active, although weakened). Vaccines may also be biochemically synthesized or made through recombinant DNA techniques.

IRB CONSIDERATIONS

The development of vaccines is of considerable benefit to society, especially in the case of devastating or highly infectious diseases. The direct benefit to the individual subject receiving a new vaccine is the possibility of immunity (*i.e.*, protection against future disease). The benefits of such immunity will vary depending on: (1) the severity of the disease to be avoided; (2) the likelihood that the subject will be exposed to the infectious disease; and (3) in the case of certain diseases, the likelihood that the subject would suffer adverse consequences should he or she contract the disease. Some populations will be at greater risk of contracting an infectious disease than others, either because they are more likely to be exposed to the disease or because they have an increased susceptibility to it. Among those who contract an infectious disease, there may be some sub-groups that are particularly vulnerable to adverse consequences (*e.g.*, children, persons of advanced age, or persons suffering from other illnesses).

For most diseases, participation in vaccine trials carries the generally small risk of contracting the disease. [In some vaccine trials (*e.g.*, HIV) there is no such risk. In the case of HIV vaccine research, the lack of risk is due to the manner in which the vaccine is derived.] The risks of participating in a vaccine trial also include adverse effects unrelated to the disease in question (*e.g.*, slight fever, headache, muscle soreness, or muscle aches). Such side effects are usually short-lived, tolerable, and not life-threatening. Again, the degree of risk associated with participating in a vaccine trial varies depending on the subjects' vulnerability to the adverse side effects of the vaccine. Some subjects may have an allergic or anaphylactic (*i.e.*, a decrease rather than an increase in immunity) reaction to the vaccine. [Anaphylactic reactions to vaccines cause the recipient to be hypersusceptible to the disease.](#) Such reactions are generally unpredictable, and may be serious or potentially life-threatening.

The IRB should be aware of other risks associated with vaccine trials, including the possibility that vaccines produced synthetically or using recombinant DNA techniques may present risks as yet unknown, that groups often most likely to benefit from receiving a vaccine are often the most vulnerable to coercion (*e.g.*, institutionalized persons or children), and that subjects in control groups may erroneously assume that they have been immunized.

When determining whether the risks are reasonable in relation to the benefits, IRBs should consider the severity of the disease, the risk of contracting the disease, and any special vulnerability of the subject population to the potential adverse effects of the vaccine. The most difficult cases are those in which the subjects most likely to benefit from participating in the vaccine trial are also the subjects at the greatest risk of suffering from the vaccine's potential adverse effects.

Anaphylactic reactions to vaccines cause the recipient to be hypersusceptible to the disease.

Review

Cytokines and autoantibodies to cytokines

K Bendtzen et al. Stem Cells. 1995 May.

[Show details](#)[Full text links](#)[Cite](#)

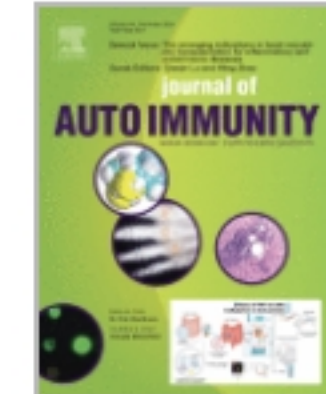
Abstract

Cytokines are essential components of our defense and repair systems but also potentially harmful mediators of infectious and immunoinflammatory reactions. Clinically important cytokines function systemically as pleiotropic hormones with overlapping effects on many cell types. All engage in a complex network of agonists and antagonists. Some immunoglobulin G (IgG) antibodies have been found to be potent and specific regulators of cytokines. These antibodies bind interleukin (IL-1)alpha, IL-6, IL-10, leukemia inhibitory factor (LIF), and interferon (IFN)-alpha/beta with exceptional force. They neutralize their corresponding cytokines ex vivo and perhaps in vivo, although they may also function as cytokine carriers. The biological role of autoantibodies to cytokines is not yet understood, but they may provide a level of regulation not appreciated at present.

Abstract



important for the regulation of B cell function (IL-4, IL-10) was also assessed. Production of TNF alpha, TNF beta and IL-10 was significantly elevated in transient hypogammaglobulinemia. The data from the ELISPOT assay suggested, that in these patients also the number of cells secreting TNF alpha after PHA stimulation was increased. These results indicate, that elevated TNF alpha production was probably due to both an enhanced release and an increased number of circulating secreting cells. As the methods employed in the quantification of cytokine levels in culture supernatants do not allow identification of the cytokine producing cell, the studies on intracellular expression of cytokines were undertaken. The fluorochrome-labelled monoclonal antibodies against the cell surface markers and a given cytokine were used simultaneously to identify the cellular source of cytokine production. The intracellular IL-4 expression in CD4+ lymphocytes from patients with transient hypogammaglobulinemia was comparable to that of control while the number of CD4+ lymphocytes expressing TNF alpha, TNF beta and IFN gamma was elevated. The number of CD14+ cells (monocytes) producing of TNF alpha was comparable to the control. These results suggest that an excessive Th-1 type response may contribute to pathology of this disease. In patients with isolated IgA



Forum

Vaccination and Autoimmunity— ‘vaccinosis’: A Dangerous Liaison?

Y Shoenfeld^{f1}, A Aron-Maor

[Show more](#) ✓

[+](#) Add to Mendeley [🔗](#) Share [🗣️](#) Cite

<https://doi.org/10.1006/jaut.1999.0346> ↗

[Get rights and content](#) ↗

Abstract

The question of a connection between vaccination and autoimmune illness (or phenomena) is surrounded by controversy. A heated debate is going on regarding the causality between vaccines, such as measles and anti-hepatitis B virus (HBV), and multiple sclerosis (MS). Brain antibodies as well as clinical symptoms have been found in patients vaccinated against those diseases. Other autoimmune illnesses have been associated with vaccinations. Tetanus toxoid, influenza vaccines, polio vaccine, and others, have been related to phenomena ranging from autoantibodies production to full-blown illness (such as rheumatoid arthritis (RA)). Conflicting data exists regarding also the connection between autism and vaccination with measles vaccine.